Brian Tang

AI Security & Privacy Researcher

https://www.bjaytang.com

Mission: Seeking a role as a Research Scientist or Product-Minded Researcher in AI Security, Safety, and Privacy. I build attacks, defenses, and products for LLMs, VLMs, and coding agents, and love turning research into real-world product impact. My expertise lies at the intersection of AI, human factors, product, and security & privacy.

Education

Ph.D. Candidate | ***Computer Science and Engineering*** — Fall 2021 – Present
University of Michigan - Ann Arbor — Advised by Kang G. Shin
Thesis: Security and Privacy Challenges with Vision-Language Models and Smart Glasses.

Bachelor of Science | ***Major: Computer Science*** — Fall 2017 – Winter 2020
University of Wisconsin - Madison — Advised by Kassem Fawaz, Varun Chandrasekaran, Somesh Jha

Experience

Research Intern — Summer 2026
IBM Research - Thomas J. Watson Research Center — Manager: Ian Molloy, Mentor: Fred Araujo

- Securing intent drift, prompt injection, secret egress, memory poisoning for sandboxed coding agents. Using reference monitors, plan-time sketches, and effect system graphs to create task-level dynamic permission scoping.
- Ran comprehensive long-horizon modified evals (e.g., SWE-Bench Pro, Terminal-Bench 2.0) on sandboxed Claude Code.

Graduate Research Assistant — Fall 2021 – Present
University of Michigan — Commitee: Kang G. Shin, Florian Schaub, Ke Sun, Stella Yu, Alex Halderman

- Creating automated data collection and annotation methods for training vision-language models to achieve above-human performance on in-the-wild text recognition tasks (>10k samples). [10]
- Designed and patented a real-time software privacy screen, Eye-Shield. Reduced attack success rates from 100% to 24.24% for images and from 77.27% to 15.91% for text, protecting against screen snooping on smartphones. [3]
- Developed various LLM web automation tools and document parsers for auditing data collection activities [12]. Deployed automated analysis of 47.2k Chrome Web Store extensions [4], 2.9k online trackers [7], and 1.4k cookie banners [1], finding many instances of misleading disclosures and non-compliance.
- Built and evaluated an LLM advertising chatbot using GPT-4o and GPT-3.5. Our user study with 179 participants found that ads influenced 13.07% more participants, with 19.05% more having positive reactions to products. Discovered that serving ads decreases LLM performance on math, reasoning, and reading comprehension tasks by 2-3%. [2]
- Served as server admin, social organizer, equipment curator, and mentoring role for the Real-Time Computing Lab.

Undergraduate Research Assistant — Fall 2018 – Spring 2021
University of Wisconsin - Madison — Advisors: Kassem Fawaz, Somesh Jha, Mentor: Varun Chandrasekaran

- Researched fairness properties of face recognition systems. [5]
- Created a controller for social robots to preserve conversational privacy. [6]
- Explored using physical invariants from LiDAR to improve ML classifier robustness against adversarial attacks. [15]
- Developed an anti face recognition system using adversarial attacks to protect online photo privacy. [8]

Software Engineering Intern — Summer 2019
Roblox Corporation — Manager: Tyler Berg

- Designed and implemented production autocomplete, smart-cursor movement, and suggestion features for Roblox Studio's script editor using TDD. Integrated 30 JavaScript Squish tests to auto-validate UI behavior and prevent errors.

Software Engineering Intern — Summer 2018
Optum, UnitedHealth Group — Manager: Kyle Erickson

- Designed and implemented an attack-surface visualization that aggregated and normalized 50M+ vulnerability and asset records, produced correlated risk scores, and an interactive dashboard. Presented results to Optum's leadership.

Publications

[1] Brian Tang, Duc Bui, and Kang G. Shin. "Navigating Cookie Consent Violations Across the Globe". **34th USENIX Security Symposium**. *Acc Rate: 17%*. 2025. [paper].

[2] Brian Tang, Kaiwen Sun, Noah T. Curran, Florian Schaub, and Kang G. Shin. "Ads that Talk Back: Implications and Perceptions of Injecting Personalized Advertising into LLM Chatbots". **Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (UbiComp/IMWUT)**. *Acc Rate: 20%*. 2025. [paper].

[3] Brian Tang and Kang G. Shin. "Eye-Shield: Real-Time Protection of Mobile Device Screen Information from Shoulder Surfing". **32nd USENIX Security Symposium**. *Acc Rate: 17%*. 2023. [paper].

[4] Duc Bui, Brian Tang, and Kang G. Shin. "Detection of Inconsistencies in Privacy Practices of Browser Extensions". **44th IEEE Symposium on Security and Privacy**. *Acc Rate: 13%*. 2023. [paper].

[5] Harrison Rosenberg, Brian Tang, Kassem Fawaz, and Somesh Jha. "Fairness Properties of Face Recognition and Obfuscation Systems". **32nd USENIX Security Symposium**. *Acc Rate: 17%*. 2023. [paper].

[6] Brian Tang, Dakota Sullivan, Bengisu Cagiltay, Varun Chandrasekaran, Kassem Fawaz, and Bilge Mutlu. "Confidant: A Privacy Controller for Social Robots". **17th ACM/IEEE International Conference on Human-Robot Interaction**. *Acc Rate: 26%*. 2022. [paper].

[7] Duc Bui, Brian Tang, and Kang G. Shin. "Do Opt-Outs Really Opt Me Out". **29th ACM Conference on Computer and Communications Security**. *Acc Rate: 18%*. 2022. [paper].

[8] Varun Chandrasekaran, Chuhan Gao, Brian Tang, Kassem Fawaz, Somesh Jha, and Suman Banerjee. "Face-Off: Adversarial Face Obfuscation". **21st Privacy Enhancing Technologies Symposium**. *Acc Rate: 22%*. 2021. [paper].

Patents

Real-Time Protection For Mobile Devices From Shoulder Surfing [3] — Spring 2023
U.S. Pat. App. No. 63/468,650-Conf. #8672 — Filed

Grants

An Efficient Real-Time Knowledge Base for Smart Glasses and Smartphones — Spring 2025
Samsung (START); Converted to joint proposal w/ Ke Sun, Anhong Guo, Kang G. Shin — Granted, $200k, 3-Pages
Ideated Project with Liangkai Liu. Wrote 1/2 of the Proposal.

I-SEE: Intelligent Vehicular Perception and Control — Spring 2025
General Motors R&D — Granted, $145k, 3-Pages
Ideated Project with Postdoc and 2 Ph.D. Students. Wrote 1/2 of the Proposal.

Evaluating Privacy and Surveillance Risks of Large Language Models — Winter 2025
National Artificial Intelligence Research Resource Pilot (NAIRR, $10k) — Granted, $10k, 3-Pages
Outlined and Scoped Projects. Wrote Full Proposal.

Securing Interactions between Driver and Vehicle Using Batteries — Summer 2023
National Science Foundation (NSF) Cloud Credits (Cloudbank) — Granted, $16k, 2-Pages
Wrote 2 Pages and Midterm Reports.

Securing Cyber-Physical System Communication and Control — Spring 2023
Defense University Research Instrumentation Program (DURIP). — Granted, $300k, 19-Pages
Initiated and Organized Proposal Structure and 25 Equipment Orders. Wrote 1/3 of the Proposal.

Preprints

[9] Brian Tang, Fred Araujo, Xiaokui Shu, and Ian Molloy. "GRAPNEL: Securing Intent Drift in Coding Agents". **In Preparation: 36th USENIX Security Symposium** (2027).

[10] Brian Tang and Kang G. Shin. "Hawkeye: Reading Illegible Text with Vision Language Models". **In Preparation: Annual Meeting of the Association for Computational Linguistics (ACL)** (2026).

[11] Noah T. Curran, Brian Tang, and Kang G. Shin. "Motion-Based Visual Instruction Tuning". **In Submission: IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)** (2026).

[12] Brian Tang and Kang G. Shin. "Steward: Natural Language Web Automation". **arXiv** (2024). [paper].

[13] Noah T. Curran, Minkyoung Cho, Ryan Feng, Liangkai Liu, Brian Tang, Pedram Mohajer Ansari, Alkim Domeke, Mert D. Pesé, and Kang G. Shin. "Short: Achieving the Safety and Security of the End-to-End AV Pipeline". **1st Cyber Security in Cars Workshop (CSCS) at CCS**. *Acc Rate: 39%*. 2024. [paper].

[14] Bulut Gozubuyuk, Brian Tang, Mert D. Pesé, and Kang G. Shin. "I Know What You Did (In Your Car) Last Summer: Privacy Implications of Android Automotive OS". **arXiv** (2024). [paper].

[15] Varun Chandrasekaran, Brian Tang, Nicolas Papernot, Kassem Fawaz, Somesh Jha, and Xi Wu. "Rearchitecting Classification Frameworks For Increased Robustness". **arXiv** (2020). [paper].

Honors and Awards

Bloomberg Summer of Puzzles Competition (*Finalist*) — Spring 2024
Puzzle Hunt Competition — Free Trip to NYC HQ

Travel Grants — 2023 – 2025
Rackham, UMich CoE, USENIX — $4k

3 Minute Thesis Competition (*Finalist*) — Fall 2023
Recovering Privacy and Autonomy in the Era of Large Language Models — 1 of 26 Finalists

College of Engineering Fellowship — Fall 2021
University of Michigan 1st year PhD fellowship — $90k

CVS Health Foundation Program — Fall 2017
Scholarship (Outstanding Children of CVS Employees) — $5k

Presentations

Implications and Perceptions of Injecting Personalized Advertising into LLM Chatbots [2] — Spring 2026
FTC, Washington D.C. | Third Federal Trade Commission Conference on Marketing and Public Policy — Event Link

CS47500 Human-Computer Interaction (Guest Lecturer) — Spring 2026
Purdue University
Taught HCI Students About AI Data Collection Studies, AI Security & Privacy User Studies (1 Hour)

Navigating Cookie Consent Violations Across The Globe [1] — Aug 2025
Seattle, WA | *USENIX Security Symposium*

Defending Against Deepfakes and Disinformation (Guest Lecturer) — Fall 2024
University of Michigan Law School — Link to Slides
Taught 30 Law Students About ML, GANs, Deepfakes, Stable Diffusion (1.5 Hours)

Steward: Natural Language Web Automation [12] — Mar 2024
Ann Arbor, MI | *SECRIT Security Reading Group*

Recovering Privacy and Autonomy in the Presence of Language Models — Sept 2023
Ann Arbor, MI | *3 Minute Thesis Finalist Competition (Engineering Graduate Symposium)*

Eye-Shield: Real-Time Protection of Mobile Device Screen Information from Shoulder Surfing [3] Aug 2023
Anaheim, CA | *USENIX Security Symposium*

Confidant: A Privacy Controller for Social Robots [6] Mar 2022
The Internet | *ACM/IEEE International Conference on Human-Robot Interaction*

Face-Off: Adversarial Face Obfuscation [8] Jan 2021
The Internet | *VMWare - NSF: Data Privacy and Edge Computing*

Face-Off: Adversarial Face Obfuscation [8] July 2021
The Internet | *Proceedings on Privacy Enhancing Technologies Symposium*

Interests

Artificial Intelligence: Adversarial ML, Computer Vision, NLP, DNNs, CNNs, (M)LLMs, Agents, VLMs, RAG
Security and Privacy: Usable Privacy, Online Privacy, Face Recognition, Social Privacy, Mobile Privacy, Surveillance
Human-Computer Interaction: User Studies, Social Robotics, Mobile Computing, Real-Time & Cyber-Physical Systems

Skills

Programming: Python, JavaScript, HTML, SQL, GLSL, C++, Kotlin, LaTeX, Linux, Bash, Godot
Software Development: GitHub, Perforce, Qt, NginX, Flask, Squish, AWS, Redis, PostgreSQL, OpenGL, d3.js, Electron
Machine Learning: TensorFlow, PyTorch, Keras, Pandas, NumPy, HuggingFace, Transformers, YOLO, Llama, PEFT
Languages: English (Native), Chinese Mandarin (Spoken-Only), Japanese (Beginner), French (Beginner)
Flight Experience: Cessna 172 – 2hrs | Cessna 152 – 2hrs
Hobbies & Interests: Reading, Hiking, Meditation, Camping, Music Production, Videogames, Tabletop RPGs

Service

External/Sub Reviewer Spring 2020 – Summer 2026
USENIX Security, PoPETS, NeurIPS, CHI, Ubicomp, HAI, IEEE Privacy 9 Papers

Poster PC Committee Member Spring 2024 – Spring 2025
IEEE S&P 2024-2025

Co-Chair/Organizer Fall 2025
Prof. Kang G. Shin's Retirement Symposium
87 Attendees, 3 Distinguished Speakers (Atul Prakash, Mingyan Liu, Farnam Jahanian)